Name: ______________________________ ________ **/ 25**

Please circle your instructor's name: Kevin Meek James Gossell Margaret Short

There are 25 points possible on this quiz. No aids (book, calculator, etc.) are permitted. **Show all work for full credit.**

1. **[10 points]** You are asked to contruct a right circular cylinder with radius r and height h such that the sum of the radius and the height is constant and equal to 6 inches, i.e.

$$r + h = 6.$$

Determine the dimensions of such a cylinder that produce the **maximum volume**. Note that the volume V of a cylinder is given by

$$V = \pi r^2 h.$$

Find the values of r and h that maximize V, and justify that those dimensions give the maximum volume.

2. [8 points] Evaluate the following limits. **Show your work**, uncluding appropriate use of limit notation. If you use L'Hôpital's rule, you must indicate where you are using it by writing $\overset{H}{=}$ or $\overset{L'H}{=}$ or something similar. Use ∞ or $-\infty$ where appropriate, and if the limit does not exist, write DNE and provide a justification.

a. $\displaystyle\lim_{t\to 2} \frac{e^{t-2}-t+1}{t^2-4t+4}$

b. $\displaystyle\lim_{x\to 0^+} 2x\ln(3x)$

3. [7 points] Evaluate the following indefinite integrals (antiderivatives):

a. $\displaystyle\int \left(\sqrt{x}+4e^x-\sin(x)\right)\,dx$

b. $\displaystyle\int \frac{2x^5-x^2+3}{x^2}\,dx$